Grammaticalization in Kinyarwanda mediated by age and gender despite ideologies about region

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Today we're talking about...

What **morphosyntactic variation** does Kinyarwanda have?

Today we're talking about...

Who says what differently?

Is there **ongoing change?**

What **facilitates** change?

Kinyarwanda

Bantu; ~10 million speakers

national lg. of Rwanda (East Africa)

agglutinative morphology

(1) Ba-aá-ra-gi-sóm-e-ye.

3pl.SBJ-DIST\PST-DJ-OBJ-DIST\read-APPL-PFV.PST

‘They were reading it for someone (yesterday or earlier).’

“disjoint” marker (DJ)

(1) Ba-aá-ra-gi-sóm-e-ye.

3pl.SBJ-DIST\PST-DJ-OBJ-DIST\read-APPL-PFV.PST

‘They were reading it for someone (yesterday or earlier).’

(Ngoboka & Zeller 2017,
van der Wal 2017)

“disjoint” marker (DJ)

(1) Ba-aá-ra-g

3pl.SBJ-DIST\PST-DJ-OBJ-

‘They were reading it for so

distribution of disjoint marker depends on… a lot!

information structure
TAM
constituency of vP
polarity
embeddedness…

(Ngoboka & Zeller 2017, van der Wal 2017, Halpert 2012, Nshemezimana & Bostoen 2017)

r 2017,
al 2017)

ra- often encodes
near future / present progressive
(but not always)

(2) Ba-ra-gi-som-er-a.

3pl.SBJ-FUT.near-OBJ-read-APPL-IPFV

‘They are about to read it.’

previous fieldwork:

(An & Umuhoza 2023, An & Ngoboka 2025)

ra- use **varies**

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(An & Umuhoza 2023, An & Ngoboka 2025)

ra- use **varies**

four "variables" selected

1 near future / present progressive *ra-* may be present or absent in: **negation**

(3) Nti-ba-(ra)-som-á.

NEG-3pl.SBJ-FUT.near-NEG\read-NEG\IPFV

‘They are not about to read.’

1 near future / present progressive *ra-* may be present or absent in: **relativization**

(4) a-b-áana ba-(ra)-som-á

AUG-2-children 3pl.SBJ-FUT.near-REL\read-REL\IPFV

‘children who are not about to read’

1 near future / present progressive *ra-* may be present or absent in: **participials**

(an embedded category selected by a few lexical heads)

(5) N-saang-a bá-(ra)-som-a.

1sg.SBJ-CJ\realize-IPFV PART\3plSBJ-FUT.near-read-IPFV

'I realize they are about to read.'

2

ra- may be obligatory or optional before ***ngo* 'that'**

(6) N-a-(ra)-vúz-e ngo haanze híijimye.

1sg.SBJ-PST-DJ-say-PFV that it's dark outside

'I said that it's dark outside.'

3

present progressive *ra-* competes with a **periphrastic**

(7) Ba-ra-som-a. Ba-ri gu-som-a.

3pl.SBJ-PRES.PROG-read-IPFV 3pl.SBJ-COP INF-read-IPFV

'They are reading.'

4

ra- encodes
near future or present progressive

(8) Ba-ra-som-a.

3pl.SBJ-**FUT.near~PRES.PROG**-read-IPFV

‘They are about to read ~ they are reading.’

1

near future / present progressive *ra-* may be present or absent in: **negation, relativization, participials**

2

ra- may be obligatory or optional before ***ngo* 'that'**

3

present progressive *ra-* competes with a **periphrastic**

4

ra- encodes **near future or present progressive**

Who do Rwandans **think** use *ra-* differently?

An & Ngoboka 2025:

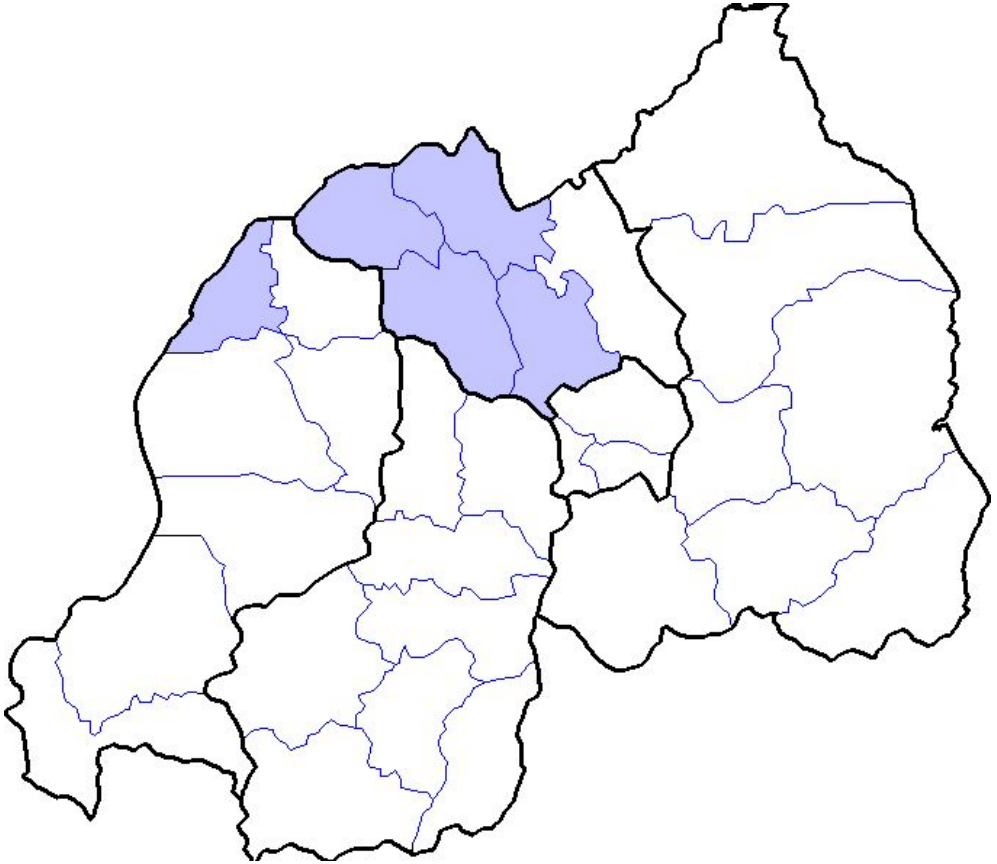
expectations / ideologies

about these variables

(N = 15)

	aware at all	*ra-* is **young**	*ra-* is **Northwestern**	*ra-* is used by **any gender**
negated *ra-* 1	13	4	3	0
ra- before *ngo* 2	11	4	1	0
periphrastic 3	13	3	1	0

Rwandans think 1 2 3 are **Northwestern** and **young**



“NW dialects”

ikirera, ikigoyi

stigmatized

Who actually **accepts** *ra-* differently?

survey of 65 respondents

ages 23-59

all five regions of Rwanda

17 self-reported users of NW dialects

Who actually **accepts** *ra-* differently?

Likert scales for sentences with / without *ra-* (White & Roberts 2022)

(worst) 1 - 5 (best)

Morphological Preference Score (MPS)

score with *ra-* minus score without *ra-*

max MPS 4 = max preference **for** *ra-*

min MPS -4 = max preference **against**

Who actually **accepts** *ra-* differently?

1

2

3

4

1

near future / present progressive *ra-*
may be present or absent in:
negation, relativization, participials

1
NEG

1
REL

1
PART

crosslinguistic evidence:
ra- is **innovative** in these environments

→ change in favor of *ra-*?

showed preference for *ra-*

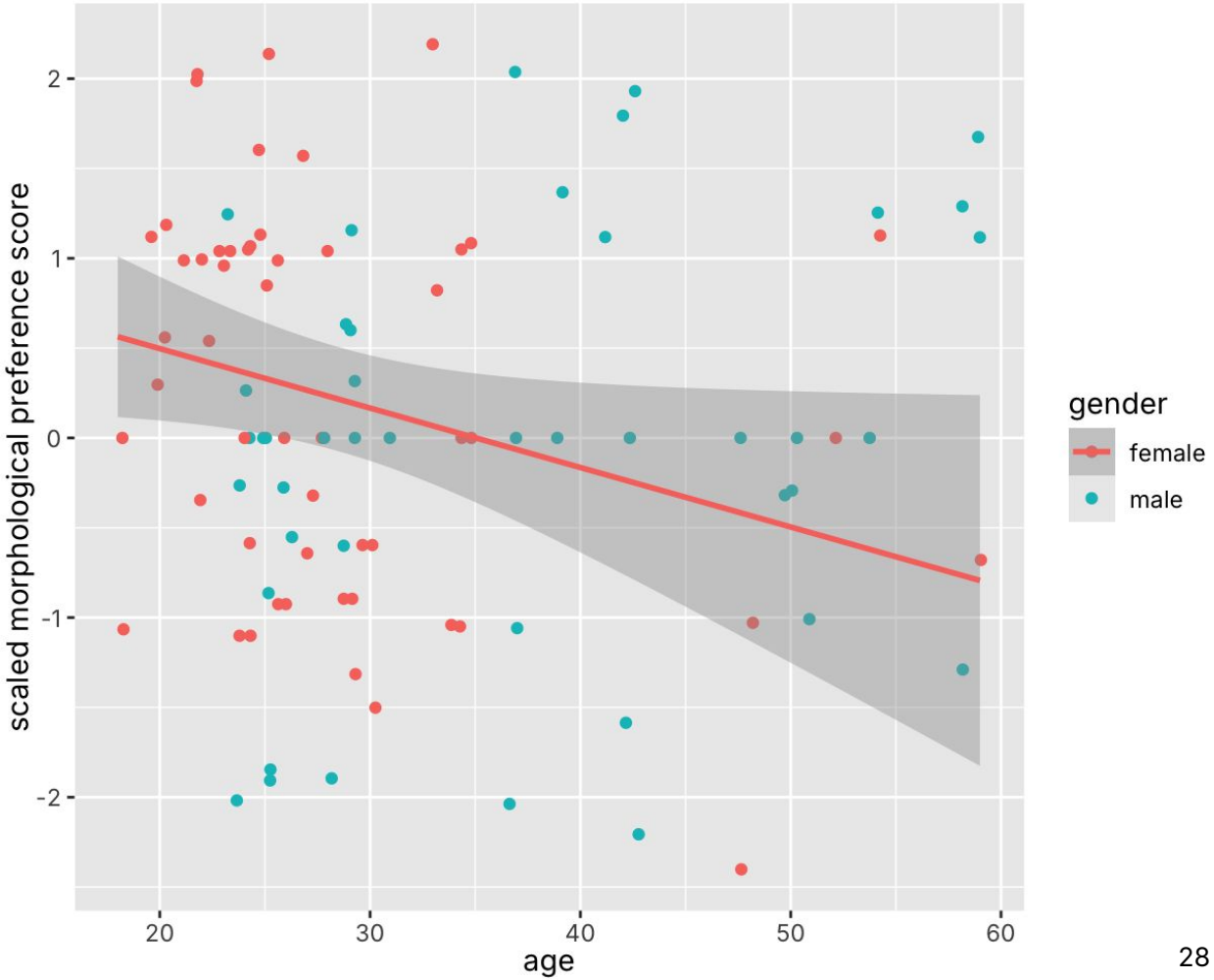
(Likert scores > 0 after scaling within participant)

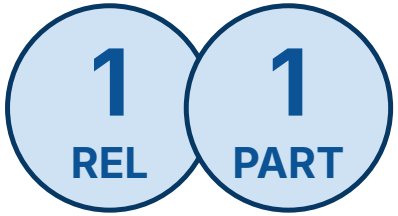
64% **61%** **41%**

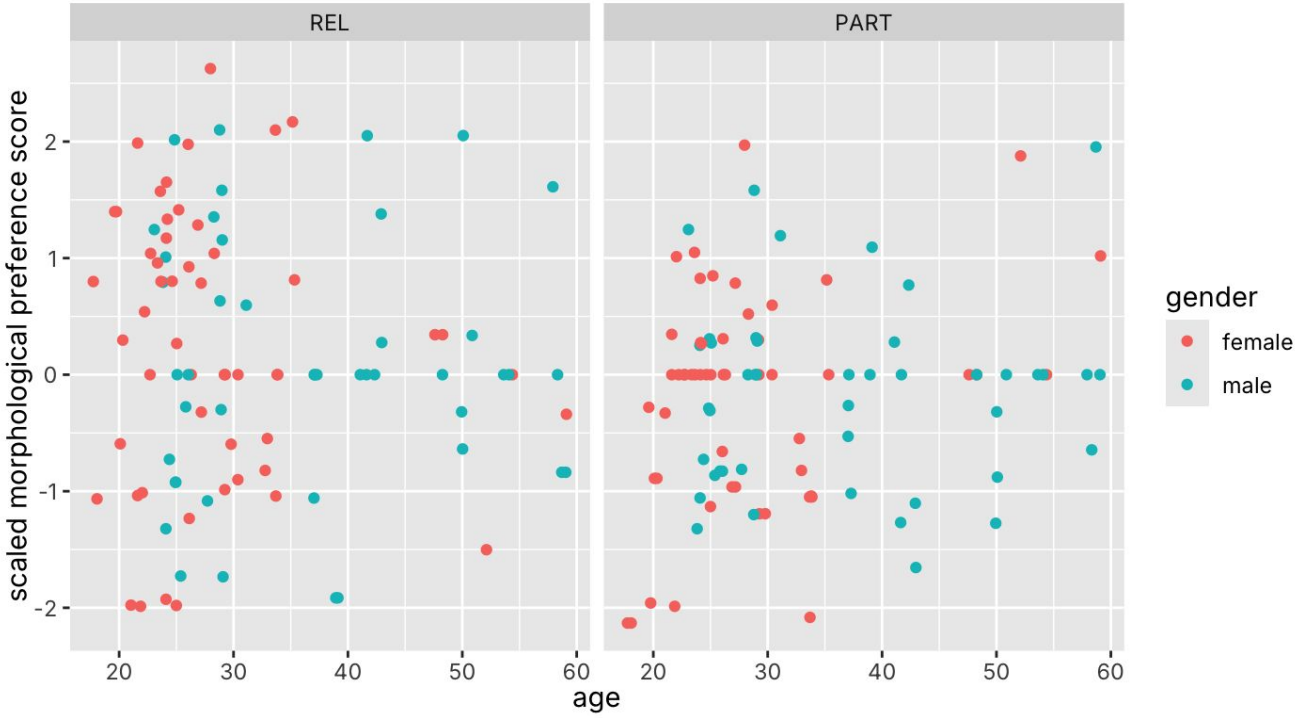
1
NEG

main effect of gender (β = -2.084, p = 0.199) attenuated by increasing age in significant interaction (β = 0.055, p - 0.038)

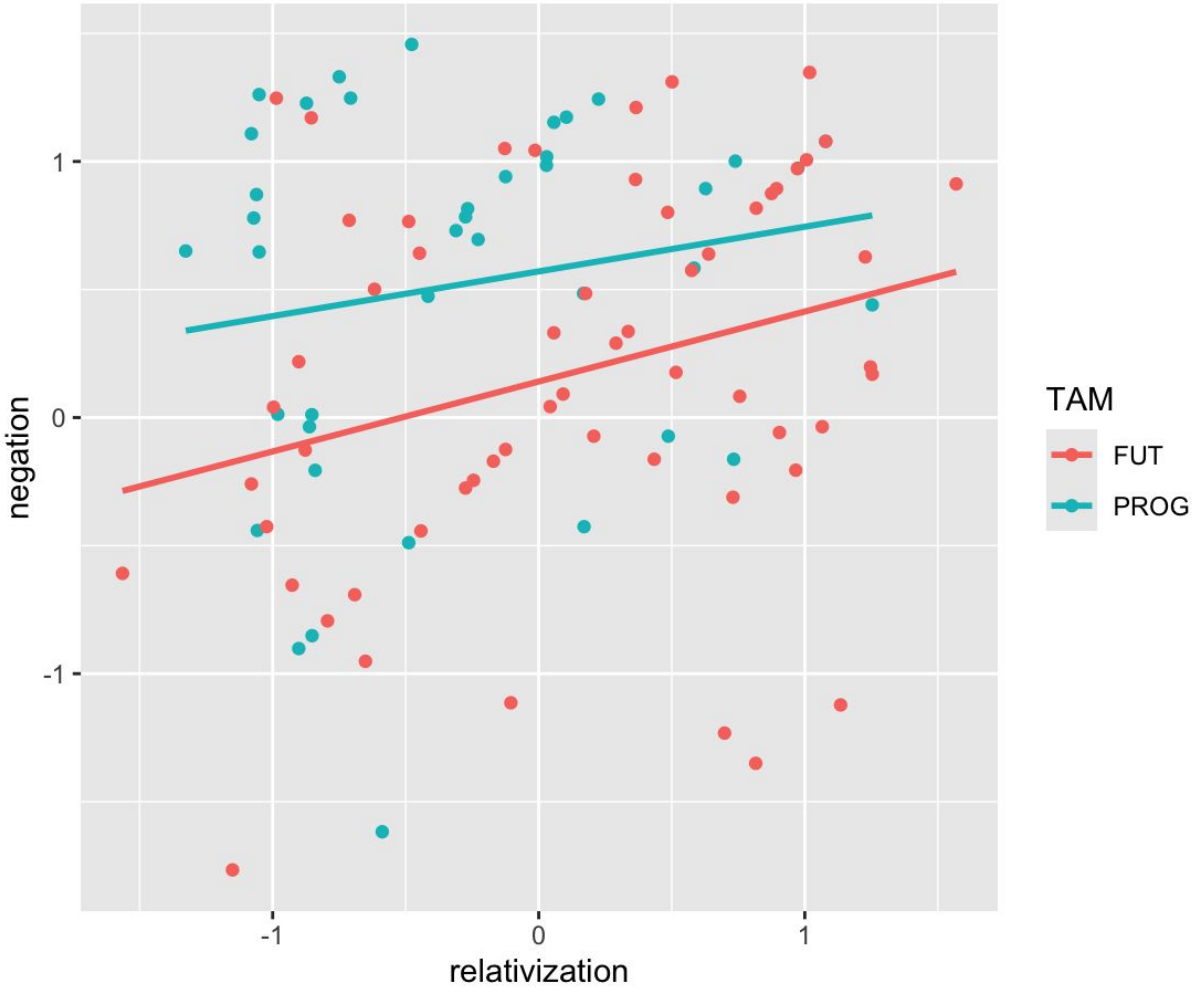
young women prefer *ra-* in negation







no effects of age, gender, region, or dialect



post-hoc:

higher scores for *ra-* in relativization predict higher scores in negation ($\beta = 0.273$, $p = 0.026$)

change in favor of *ra-* led by young women

stable variation?

64% **61%** **41%**

2

ra- is obligatory or optional before **ngo ‘that’**

2

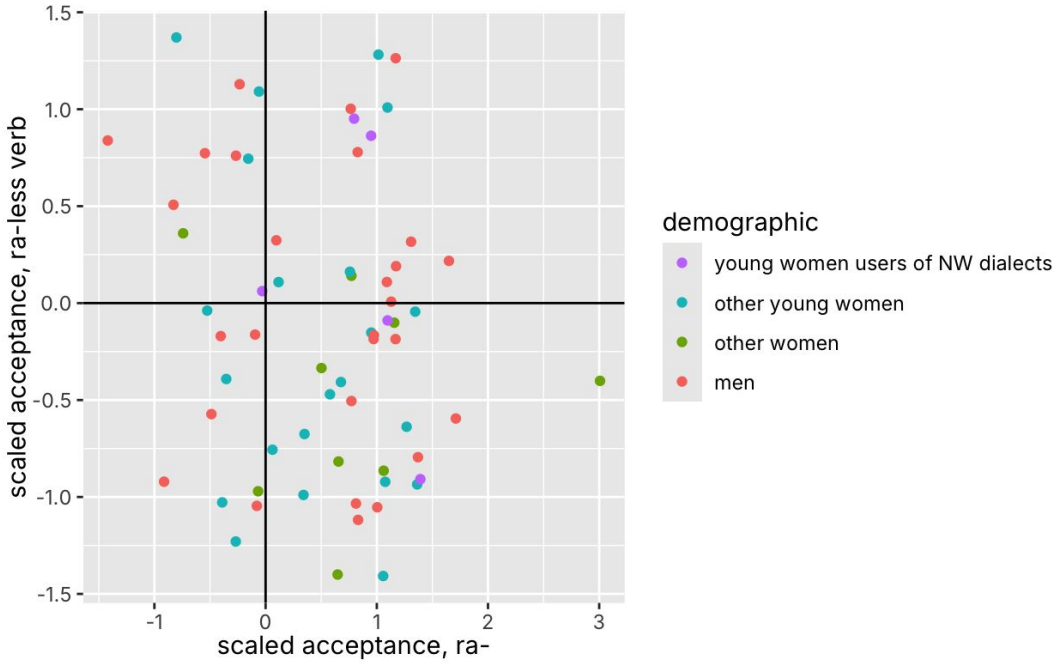
women
(β = -1.449, p = 0.060)

young women
(β = 0.046, p = 0.039)

young women users of NW dialects
(β = -0.09, p = 0.034)

all rate *ra*- higher

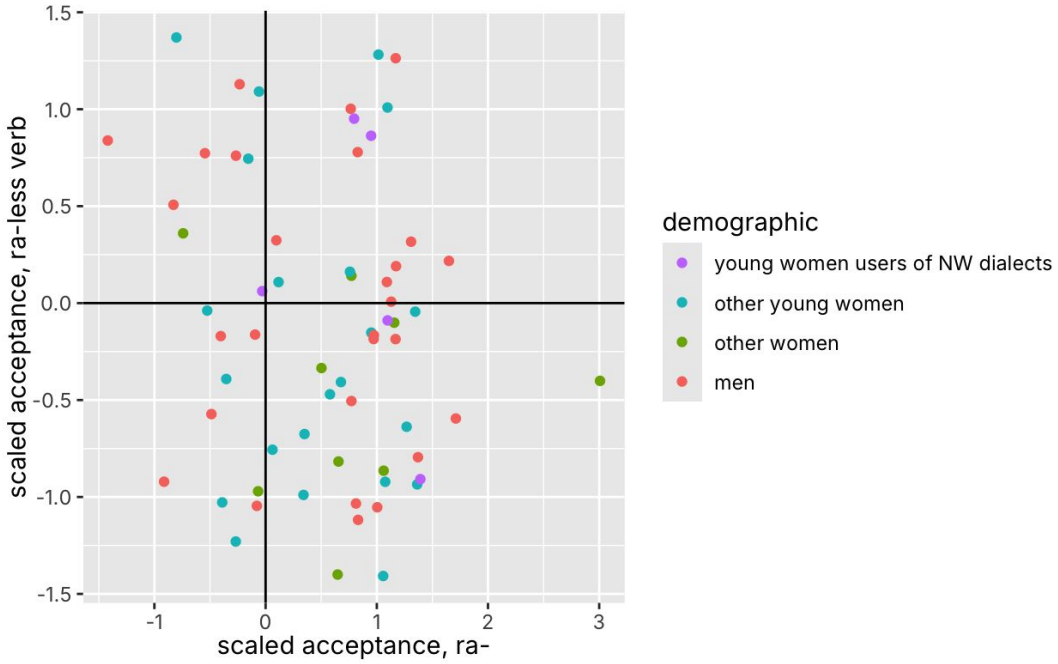
no patterns re: *ra*-less counterparts



2

optional *ra-* becoming **more acceptable** but not more obligatory

change led by **young women who use NW dialects**



1

near future / present progressive *ra-* may be present or absent in: **negation, relativization, participials**

young women, *ra- in* negation only

2

ra- may be obligatory or optional before ***ngo* 'that'**

young women users of NW dialects, *ra-* more preferable

3

present progressive *ra-* competes with a **periphrastic**

4

ra- encodes **near future or present progressive**

1

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present progressive *ra-* competes with a **periphrastic**

4

ra- encodes **near future or present progressive**

no effects of age, gender, region, or dialect

Discussion

Rwandans think 1 2 3 are **Northwestern** and **young**

role of Northwest?

no clear pattern

sporadic effects, small sample size, stigmatization → underreporting

In the IND-DP frame there was a significant interaction between gender and region (β = 5.962, p = 0.046) such that male Northwesterners preferred periphrastics; the interaction was attenuated by increasing age in a significant interaction (β = -0.181, p = 0.027) such that young male Northwesterners preferred periphrastics. In the REL frame there was a marginal main effect of gender (β = 1.186, p = 0.098) such that men preferred periphrastics; this effect was attenuated by region in a marginally significant interaction (β = -5.514, p = 0.091) such that male Northwesterners dispreferred periphrastics. In the PTCP frame there was a significant main effect of gender (β = -2.185, p = 0.025) such that men dispreferred periphrastics. This effect was attenuated by increasing age (β = 0.058, p = 0.042) such that young male Northwesterners preferred periphrastics

evidence for change
led by young women

evidence for **stability**

all three crosslinguistically innovative, but...

change

1
NEG

stability

1
REL

1
PART

proposal one:

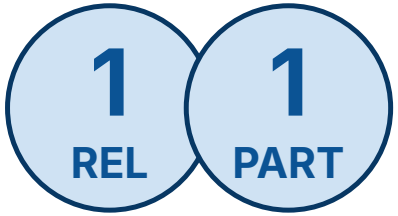
later change?
slower change?
→ visible change

early change?
faster change?
→ stability

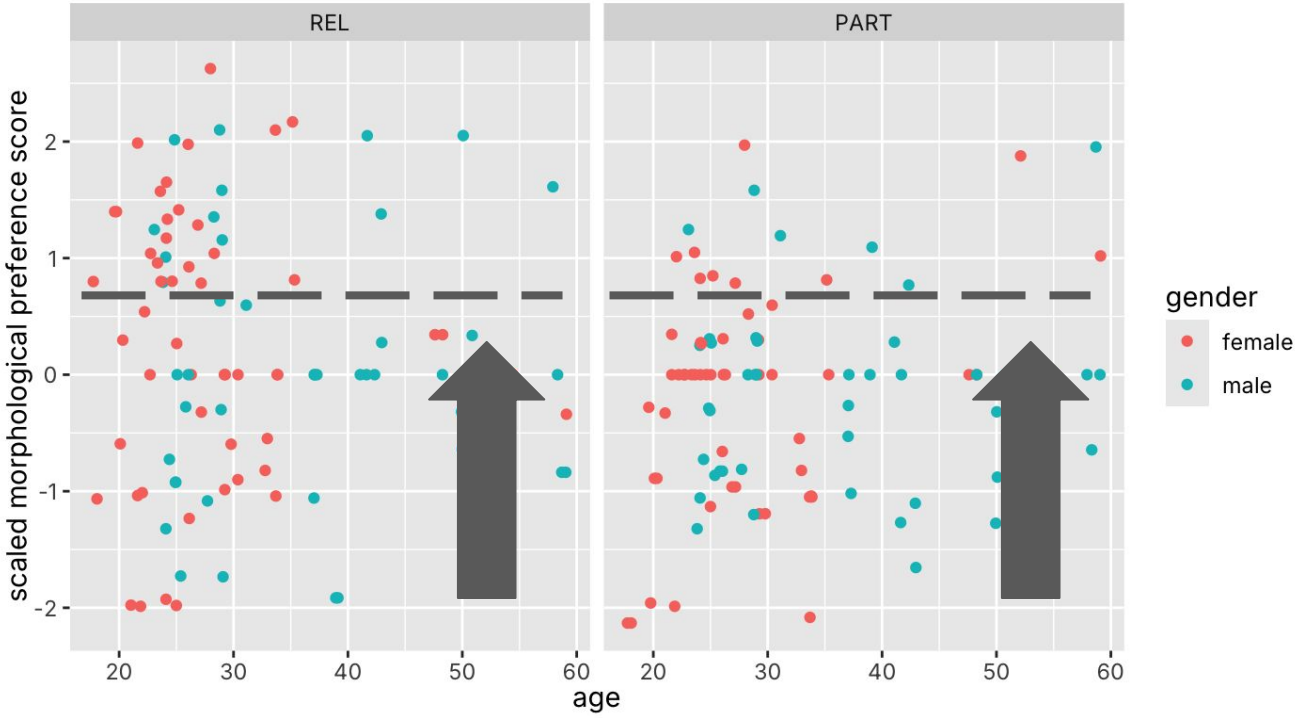
proposal two:

community change

adults in communities undergoing change may “participate in the change themselves” (Sankoff 2019)



(idealized lines!)



raised acceptance among older people removes age effects

both proposals:
more susceptible
to change. why?

(9a) unmarked *barasoma*

original distribution
(near-universal acceptance)

(9a) unmarked	*barasoma*
(9b) REL	*ba(ra)somá*
(9c) PART	*bá(ra)soma*

surface similarity
facilitates analogy

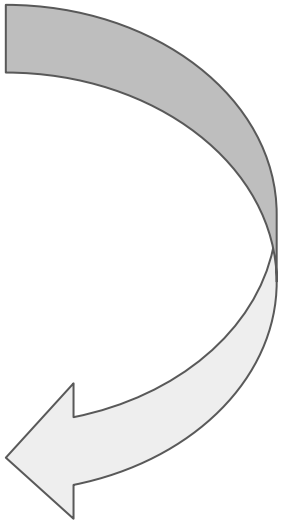
(9a) unmarked	*barasoma*
(9b) REL	*ba(ra)somá*
(9c) PART	*bá(ra)soma*
(9d) NEG	***nti**ba(ra)somá*

extra segments hinder analogy

Periphrastic negation uses relativization

(10a) *Ntaabwo* *ba(ra)somá.*
NEG they.are.about.to.read.**REL**
'They are not about to read.'
(early change?)

(10b) *Nti-ba(ra)somá.*
NEG-they.are.about.to.read
'They are not about to read.'
(late change?)



additional pressure created by REL?

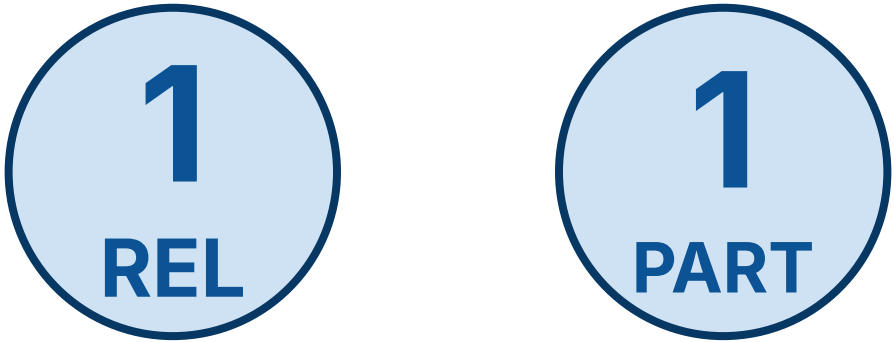
surface similarity facilitates analogical morphological change

→ currently testing this!
how? ask me / see handout!

why did one “settle” higher than the other?

61% **41%**

frequent infrequent



frequency facilitates **leveling** of competing morphosyntactic strategies (Bybee 1995, Smith 2001)

maybe:

frequency facilitates **greater use** if variation reaches stability?

wrapping up

Rwandans use *ra-*
differently!

wrapping up

some *ra-* variation is **stable**

wrapping up

ra- is **expanding into new syntactic environments**

wrapping up

1

surface similarity facilitates analogical change?

wrapping up

1

frequency facilitates higher use in stable variation?

Murakoze!

‘Thank you!’

An, Adam & Jean Paul Ngoboka. 2025. Variation and Awareness in Kinyarwanda Morphosyntax. Presented at the 56th Annual Conference on African Linguistics.
An, Adam & Solange Umuhoza. 2023. Grammaticalization in progress in Kinyarwanda. Poster, LSA Annual Meeting 2023.
Bybee, Joan. 1995. Regular morphology and the lexicon. Language and Cognitive Processes 10(5). 425–455. https://doi.org/10.1080/01690969508407111.
Halpert, Claire. 2012. Argument licensing and agreement in Zulu.
Ngoboka, Jean Paul & Jochen Zeller. 2017. The conjoint/disjoint alternation in Kinyarwanda. In Jenneke Wal & Larry M. Hyman (eds.), The Conjoint/Disjoint Alternation in Bantu, 350–389. De Gruyter. https://doi.org/10.1515/9783110490831-013.
Nshemezimana, Ernest & Koen Bostoen. 2017. The conjoint/disjoint alternation in Kirundi (JD62): A case for its abolition. In Jenneke van der Wal & Larry M. Hyman (eds.), The Conjoint/Disjoint Alternation in Bantu, 390–425. De Gruyter. https://doi.org/10.1515/9783110490831-014.
Sankoff, Gillian. 2019. Building on Empirical Foundations: Individual and Community Change in Real Time. In New Directions for Historical Linguistics. BRILL. https://doi.org/10.1163/9789004414075.
Smith, K. Aaron. 2001. The role of frequency in the specialization of the English anterior. In Joan L. Bybee & Paul J. Hopper (eds.), Typological Studies in Language, vol. 45, 361–382. Amsterdam: John Benjamins Publishing Company. https://doi.org/10.1075/tsl.45.18smi.
Wal, Jenneke van der. 2017. What is the conjoint/disjoint alternation? Parameters of crosslinguistic variation. In Jenneke van der Wal & Larry M. Hyman (eds.), The Conjoint/Disjoint Alternation in Bantu, 14–60. De Gruyter. https://doi.org/10.1515/9783110490831-002.
White, Yosiane & Gareth Roberts. 2022. Variability in speaker expectations of morphosyntactic mutation in Welsh. Glossa: a journal of general linguistics 7(1). https://doi.org/10.16995/glossa.8730.

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